NUCLEAR HAMILTONIAN

Let us consider a prototypal nuclear Hamiltonian of the kind

$$H = -\sum_{i=1}^{A} \frac{\nabla_i^2}{2m_N} + \sum_{i<j} \hat{v}_{ij}$$

Here, the two-body potential is the sum of a purely central term and a spin-dependent one:

$$\hat{v}_{ij} = v^c(r_{ij}) + v^\sigma(r_{ij})\, \vec{\sigma}_i \cdot \vec{\sigma}_j$$

With $\vec{\sigma}_i = \sigma_i^x, \sigma_i^y, \sigma_i^z$ we denote the standard Pauli matrix. The radial functions $v^c(r_{ij})$ and $v^\sigma(r_{ij})$ are determined before-hand by fitting two-body observables.

In a particular framework, pionless effective field theory, they are simply given by:

$$v^c(r) \cong C_1\, e^{-\frac{r^2}{\Lambda^2}} \qquad v^\sigma(r) \cong C_\sigma\, e^{-\frac{r^2}{\Lambda^2}}$$

where Λ is a regulator and C_1 and C_σ are low-energy constants, or parameters, that are fixed on Nucleon-Nucleon scattering data.

For a more detailed discussion on this topic, see this paper

https://arxiv.org/pdf/1412.3081.pdf

WAVE FUNCTION IN THE AFMC

$\langle RS | \Phi \rangle$ where $|\Phi\rangle = A\left[|p\uparrow\rangle, |p\downarrow\rangle, |m\uparrow\rangle, |m\downarrow\rangle \right]$

$$\underbrace{\langle RS|}_{\text{state 1}}\Phi\rangle = \det \begin{array}{c} \text{state 1} \\ \text{state 2} \\ \text{state 3} \\ \text{state 4} \end{array} \begin{bmatrix} \langle s_1 | p\uparrow\rangle_1 & \langle s_2 | p\uparrow\rangle_2 & \langle s_3 | p\uparrow\rangle_3 & \langle s_4 | p\uparrow\rangle_4 \\ \langle s_1 | p\downarrow\rangle_1 & \langle s_2 | p\downarrow\rangle_2 & \langle s_3 | p\downarrow\rangle_3 & \langle s_4 | p\downarrow\rangle_4 \\ \langle s_1 | m\uparrow\rangle_1 & \langle s_2 | m\uparrow\rangle_2 & \langle s_3 | m\uparrow\rangle_3 & \langle s_4 | m\uparrow\rangle_4 \\ \langle s_1 | m\downarrow\rangle_1 & \langle s_2 | m\downarrow\rangle_2 & \langle s_3 | m\downarrow\rangle_3 & \langle s_4 | m\downarrow\rangle_4 \end{bmatrix}$$

particle 1 particle 2 particle 3 particle 4

$|s\rangle = |s_1\rangle \otimes |s_2\rangle \otimes |s_3\rangle \otimes |s_4\rangle$

$\rightarrow$ ph (4, mpart, mpart)

$$|p\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |p\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad |m\uparrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad |m\downarrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$|\Phi\rangle = \begin{bmatrix} \begin{pmatrix}1\\0\\0\\0\end{pmatrix}_1 & \begin{pmatrix}1\\0\\0\\0\end{pmatrix}_2 & \begin{pmatrix}1\\0\\0\\0\end{pmatrix}_3 & \begin{pmatrix}1\\0\\0\\0\end{pmatrix}_4 \\ \begin{pmatrix}0\\1\\0\\0\end{pmatrix}_1 & \vdots & \vdots & \vdots \\ \begin{pmatrix}0\\0\\1\\0\end{pmatrix}_1 & \vdots & \vdots & \vdots \\ \begin{pmatrix}0\\0\\0\\1\end{pmatrix}_1 & \begin{pmatrix}0\\0\\0\\1\end{pmatrix}_2 & \begin{pmatrix}0\\0\\0\\1\end{pmatrix}_3 & \begin{pmatrix}0\\0\\0\\1\end{pmatrix}_4 \end{bmatrix}$$

$\rho_{AB}(\text{mpu}_{A,4}, \text{mpu}_{B})$

$$\langle S|\Phi\rangle \quad \begin{bmatrix} (C_1^{\uparrow}, C_1^{\downarrow}, C_1^{\uparrow}, C_1^{\downarrow}) \begin{pmatrix}1\\0\\0\\0\end{pmatrix} & (C_2^{\uparrow}, C_2^{\downarrow}, C_2^{\uparrow}, C_2^{\downarrow}) \begin{pmatrix}1\\0\\0\\0\end{pmatrix}_2 \\ (C_1^{\uparrow}, C_1^{\downarrow}, C_1^{\uparrow}, C_1^{\downarrow}) \begin{pmatrix} \\ \\ \\ \end{pmatrix} & \vdots \\ \vdots & \end{bmatrix}$$

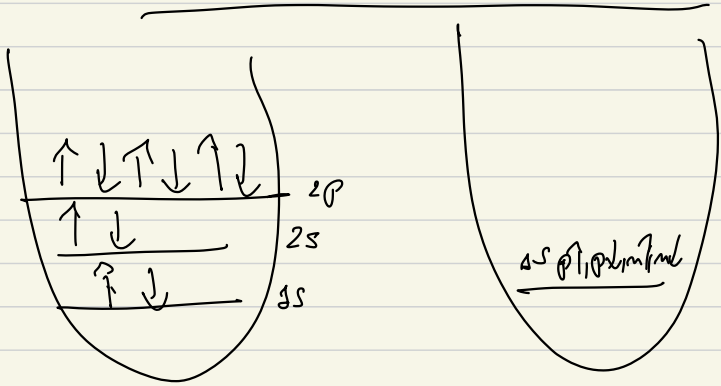
$$|s_1\rangle = \left(c_{p\uparrow},\ c_{p\downarrow},\ c_{m\uparrow},\ c_{m\downarrow} \right)$$

$$\langle s|\Phi\rangle = \det \begin{bmatrix} c_1^{p\uparrow} & c_2^{p\uparrow} & c_3^{p\uparrow} & c_4^{p\uparrow} \\ c_1^{p\downarrow} & c_2^{p\downarrow} & c_3^{p\downarrow} & c_4^{p\downarrow} \\ c_1^{m\uparrow} & c_2^{m\uparrow} & c_3^{m\uparrow} & c_4^{m\uparrow} \\ c_1^{m\downarrow} & c_2^{m\downarrow} & c_3^{m\downarrow} & c_4^{m\downarrow} \end{bmatrix} = \det$$

$$|{}^2H\rangle = A\left[|p\uparrow\rangle |m\uparrow\rangle \right]$$

$$|{}^3H\rangle = A\left[|p\uparrow\rangle, |m\uparrow\rangle, |m\downarrow\rangle \right]$$

$$|{}^4He\rangle = A\left[|p\uparrow\rangle, |p\downarrow\rangle, |m\uparrow\rangle, |m\downarrow\rangle \right]$$



$$|\Phi_{He}\rangle = A\left[\Phi_{1s}\,|p\uparrow\rangle\,\Phi_{1s}\,|p\downarrow\rangle\,\Phi_{1s}\,|n\uparrow\rangle\,\Phi_{1s}\,|n\downarrow\rangle\right]$$

$$\Phi_{n,l,l_z}(\vec{r}) = R_{nl}(|\vec{r}|)\,Y_{ll_z}(\hat{r})$$

$$\Phi_{1s}(\vec{r}) = R_{1s}(|\vec{r}|)\cdot 1$$

$$R_{1s}(|r|) = e^{-\alpha r^2}$$ α is a variational parameter

$$\rightarrow \quad H_{WS}\,\varphi_1(r_1) = \varepsilon\,\varphi(r_1) \quad \rightarrow \text{Sh. eq 1-body}$$

$$\langle RS|\Phi\rangle = \det\begin{bmatrix} \Phi_{1s}(r_1)\langle s_1|p\uparrow\rangle & \Phi_{1s}(r_2)\langle s_2|p\uparrow\rangle & \cdots \\ \Phi_{1s}(r_1)\langle s_1|p\downarrow\rangle & \Phi_{1s}(r_2)\langle s_2|p\downarrow\rangle & \cdots \\ \Phi_{1s}(r_1)\langle s_1|n\downarrow\rangle & \Phi_{1s}(r_2)\langle s_2|n\downarrow\rangle & \cdots \quad \Phi_{1s}(r_4)\langle s_4|n\downarrow\rangle \end{bmatrix}$$

$$|\Phi\rangle = \boxed{\sum_{is,it} \psi(i_s,i_t)\,|i_s,i_t\rangle}$$

$$= \boxed{\psi_{\uparrow\uparrow\uparrow\uparrow,pnpn}}\,|\uparrow\uparrow\uparrow\uparrow,pnpn\rangle + \psi_{\uparrow\uparrow\uparrow\uparrow,pnpn}\,|$$

$\Phi_{1s}(r_1)\ldots$

do is

do it

$$\psi(is, it) \overset{?}{=} \quad \Phi_{AFDMC}$$

$$\psi(is, it) =$$